Signs of Early Sub-Clinical Atherosclerosis in Childhood Cancer Survivors

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ABSTRACT

Background. Cancer is the leading cause of death by disease among U.S. children; however, over the last two decades there has been a dramatic increase in survival in these individuals. Although free of cancer, childhood cancer survivors (CCS) are faced with a variety of chronic health problems including an increased risk of cardiovascular and metabolic abnormalities. The effect of cancer treatments on vascular structure and function in CCS has not been examined in childhood.

Methods. Measures of carotid artery stiffness (compliance and distensibility) and intima-media thickness (cIMT), brachial artery endothelial-dependent dilation (EDD), and endothelial-independent dilation (EID) were obtained from ultrasound imaging in 319 CCS who were ≥5 years from diagnosis and 208 siblings who had never been diagnosed with cancer. Participants were 9-18 years of age at examination.

Results. CCS were shorter and had greater percent body fat and lower lean body mass compared to controls. Carotid diameter (DD) and cross-sectional distensibility (CSD) was lower in CCS compared to controls. Survivors of leukemia had lower DD, CSD and compliance (CSC), indicating increased arterial stiffness, compared to controls. CSD was also lower in survivors of central nervous system (CNS) tumors than controls. There were no significant differences in measures of carotid stiffness or EDD in survivors of solid tumors compared to controls. Carotid IMT was greater in survivors with CNS tumors compared to leukemia survivors. EDD was lower in leukemia survivors than in controls, and EID was greater in survivors of CNS tumors than in controls.

Conclusion. These results demonstrate that early in life, CCS have arterial changes indicating increased risk for premature atherosclerosis and cardiovascular disease. Therefore, it is reasonable to advocate that efforts should be directed at monitoring and managing cardiovascular risk factors in CCS.

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BACKGROUND

- Cancer is the leading cause of death by disease among U.S. children ages 1 to 14 yrs[1].
- Due to advances in treatment[2,3] the 5 yr survival rate has increased from 58.1% in 1975-1977 to 83.1% in 2003-2009[1].
- Although free of cancer, CCS are now at an increased risk of cardiovascular and related metabolic abnormalities.[4]
- The primary objective of this study was to evaluate measures of vascular structure and function in a large population of CCS, while they were still children and compare them to a control group of healthy sibling children.

METHODS

- Three-hundred nineteen CCS who were ≥5 yrs from diagnosis and 208 siblings who had never been diagnosed with cancer. Participants were 9-18 yrs of age at examination (Table 1).
- Measures of carotid artery stiffness (compliance and distensibility) and intima-media thickness (cIMT), and brachial artery measures of endothelial-dependent dilation (EDD), and endothelial-independent dilation (EID) were obtained via high-resolution ultrasound imaging (Acuson, Sequoia 512, Siemens).

RESULTS

- Although CCS were slightly older, pubertal development was not different than controls (Table 1).
- There were no significant differences between CCS and controls for weight or BMI; however, CCS were significantly shorter and had significantly higher percent fat mass and lower lean body mass (Table 1).
- Survivors of leukemia had lower DD, CSD, CSC and diameter compliance (DC) compared to controls (Table 2).
- DD and CSD was lower in survivors of CNS tumors than controls (Table 2).
- Measures of carotid stiffness measures and EDD were not significantly different between controls and survivors of solid tumors (Table 2).
- EDD was lower in leukemia survivors than controls (Table 2).
- EID was greater in survivors of CNS tumors than controls (Table 2).

TABLE 1. Demographic Characteristics (Mean±SE)

			CCS by Diagnosis		
	Control (n=208)	All CCS (n=319)	Leukemia (n=110)	CNS (n=82)	Solid tumors (n=127)
Age (yrs)	13.6±0.2	14.6±0.1[a]	14.4±0.2[a]	14.7±0.3[a]	14.6±0.2[a]
Sex (M/F)	112/96	171/148	65/45	34/48	72/55
Tanner Stage	3.4±0.1	3.6±0.1	3.5±0.1	3.7±0.1	3.6±0.1
Height (cm)	159.9±0.7	158.2±0.6[a]	156.4±1.0[a,b]	156.2±1.0[a]	158.7±0.9
Weight (kg)	57.1±1.2	57.2±1.1	55.4±1.7	58.1±2.0	55.4±1.4
BMI (kg/m^2)	21.8±0.4	22.4±0.3	22.2±0.5	23.1±0.6	21.8±0.5
%Fat Mass	25.9±0.9	28.1±0.8[a]	28.6±1.1[a]	29.9±1.2[a]	26.3±1.4
LBM (kg)	39.9±0.9	38.4±0.5[a]	37.0±0.9[a]	37.3±0.9	38.6±0.8

CNS, survivors of central nervous system tumors; CCS, childhood cancer survivor; BMI, body mass index; LBM, lean body mass. Measures adjusted for age-at-study, sex, race, and Tanner stage. [a]Denotes P<0.05 for comparison to control group; [b]Denotes P<0.05 for comparison to survivors of solid tumors.

Table 2. Comparison of Carotid and Brachial Artery Measures between CCS Groups and Sibling Controls (Mean±SE)

			CCS by Diagnosis		
	Control (n=208)	All CCS (n=319)	Leukemia (n=110)	CNS (n=82)	Solid tumors (n=127)
DD	15.10±0.26	14.32±0.21[a]	14.02±0.34[a]	14.30±0.35[a]	14.61±0.28
CSC	0.63±0.02	0.60±0.01	0.59±0.02[a]	0.61±0.02	0.61±0.01
DC	1.65±0.04	1.61±0.03	1.55±0.05[a]	1.65±0.05	1.62±0.04
CSD	32.70±0.61	30.72±0.48[a]	30.09±0.79[a]	30.77±0.79[a]	31.24±0.67
cIMT	0.44±0.003	0.44±0.003	0.44±0.004	0.45±0.007[b]	0.44±0.003
EDD	8.20±0.36	7.60±0.28[a]	7.55±0.38[a]	8.13±0.47	7.65±0.37
EID	25.06±0.57	26.17±0.44[a]	26.05±0.63	26.68±0.68[a]	25.67±0.61

DD, diameter distensibility, %; CSD, cross-sectional distensibility, %; DC, diameter compliance, mm/mmHg; CSC, cross-sectional compliance, mm^{2}mmHg; cIMT, carotid intima-media thickness, mm; EDD, endothelial-dependent dilation, %; EID, endothelial-independent dilation, %. Measures are adjusted for age-at-study, sex, race, Tanner stage and percent fat mass. [a]Denotes P<0.05 for comparison to controls; [b]Denotes P<0.05 for comparison to leukemia survivors.

CONCLUSION

- These results demonstrate that early in life, childhood cancer survivors have arterial changes indicating increased risk for premature atherosclerosis and cardiovascular disease.
- Time since diagnosis had no effect on any outcome variable, which further suggests that the declines in vascular structure and function are likely established early in the course of survivorship.
- It is reasonable to advocate that efforts should be directed at monitoring and managing cardiovascular risk factors in childhood cancer survivors.